Instability #21: Proof at 90% and the Exact Remaining Gap

Structure Derived, $n = 2^{d_s}$ Refuted, $\mathrm{T}_{\mathrm{avg}}(M)$ Precision Is the Last Step

For $n = 8$, $d_s = 3$, V20.3: bracket $= \kappa/\ln 2$ to 0.23%. One input remains to derive.

Ougit Jarittum

March 2026

Abstract

This document consolidates all findings on Instability #21 following the $n = 4$ empirical test and the self-consistency analysis.

Key findings: (1) $n = 2^{d_s}$ is empirically refuted: $n = 4$ gives $d_s \approx 3.58$, not 2. The formula $|\beta| = d_s/(2\ln 2)$ does not hold universally across n. (2) The proof structure is 90% complete. All components are derived except the precise value of $T_{\mathrm{avg}}(M)$ in the mature simulation regime. (3) The bracket identity reduces to a self-consistency equation:

$$\frac{\epsilon_0^2}{2}\left(1 - \frac{1}{n}\right)\langle \Sigma_{\mathrm{inv}} \cdot T\rangle_{\mathrm{eff}} = \kappa \cdot \frac{1/\alpha - \ln n \cdot \langle 1/(1+M)\rangle_{\mathrm{ev}}}{\langle \ln(1+M)\rangle_{\mathrm{ev}}}$$

where all quantities except $T_{\mathrm{avg}}(M)$ in the p_{ev} distribution are from the δ-chain. The early-time Gamma fit for $T_{\mathrm{avg}}(M)$ gives 10% error. The mature-regime $T_{\mathrm{avg}}(M)$ would close the proof.

Contents

1 Proof Components: What Is Derived

Derived: All Structural Components

1. $f_{12} = d_s/2 = 3/2$ from Kuramoto sub-threshold ($K_{\text{eff}} < K_c$).
2. $H_{\text{after}} = \ln n - D_{\text{KL}}(\bar{P}_{\text{nb}} \| U)$ from Born rule + locking fraction ρ.
3. $T_{\text{avg}}(M)$ structural form $= 1/P(D_{\text{KL}} > K^* \sqrt{M})$ from θ-formula.
4. $p_{\text{ev}}(M_k) \propto \text{Poi}(\lambda, k)/T_{\text{avg}}(M_k)$ from mass growth dynamics ($M = 1 + \kappa k$).
5. Bracket structure: bracket $= H_{\text{after}}\kappa/(1+M) + \delta H \ln(1+M)$.
6. $\delta H \approx \delta \bar{H}$ (constant, CV 6%) from mutual cancellation of $\Sigma_{\text{inv}}(M) \cdot T_{\text{avg}}(M)$ trends.

2 The $n = 4$ Test: Refutation and Implications

Result: $n = 4$ in V20.3 (2 seeds, 120 sweeps)

n	d_s	$\lvert\beta\rvert$	$\lvert\beta\rvert/d_s$	$1/(2\ln 2)$	formula?
8	3.03 ± 0.08	2.212	0.730	0.721	$\approx$ (1.2%)
4	3.58 ± 0.10	1.847	0.516	0.721	$\times$ (28%)

$n = 4$ gives $d_s \approx 3.58$, not $\log_2 4 = 2$.
$|\beta|/d_s = 0.516 \neq 1/(2\ln 2) = 0.721$ at $n = 4$.

Confirmed: $n = 2^{d_s}$ is Refuted

d_s is determined by X-space geometry (3D), not by n. Changing n from 8 to 4 shifts d_s from 3.03 to 3.58: it increases because lower-dimensional P-space creates sparser KL graphs. It does *not* decrease to 2 as the Saturation Identity predicts. Consequence: $|\beta| = d_s/(2\ln 2)$ holds for $n = 8$ specifically, because $f_{12} = 3/2$ (Kuramoto) coincides with $d_s/2 \approx 3/2$ ($d_s \approx 3$). At $n = 4$, $d_s \approx 3.58$, so $d_s/2 \approx 1.79 \neq f_{12} = 1.50$, and the formula breaks.

3 The Self-Consistency Equation

The bracket identity bracket $= \kappa/\ln 2$ is equivalent to:

$$\underbrace{\ln n \cdot \langle 1/(1+M) \rangle_{\text{ev}}}_{\text{Term A}/\kappa} + \underbrace{\delta\bar{H}/\kappa \cdot \langle \ln(1+M) \rangle_{\text{ev}}}_{\text{Term B}/\kappa} = \frac{1}{\ln 2}$$

Substituting $\delta\bar{H} = (\epsilon_0^2/2)(1 - 1/n)\langle \Sigma T \rangle_{\text{eff}}$:

Self-Consistency Equation

$$\underbrace{\frac{\epsilon_0^2}{2}\left(1-\frac{1}{n}\right)\langle\Sigma_{\text{inv}}\cdot T\rangle_{\text{eff}}}_{\delta\bar{H}} = \kappa\cdot\frac{1/\ln 2 - \ln n\cdot\langle 1/(1+M)\rangle_{\text{ev}}}{\langle\ln(1+M)\rangle_{\text{ev}}}$$

where $\Sigma_{\text{inv}} = \sum_k P_k^{-1}$ and $\langle\cdot\rangle_{\text{eff}}$ is a specific weighted average over inter-event periods.

All symbols are from the δ-chain. The equation expresses a *detailed balance* of the CRF cascade: the entropy decay between events exactly compensates the geometric and mass contributions to bracket.

Numerically at $n = 8$: LHS $= 0.1132$ ($\delta\bar{H}$ measured), RHS $= 0.1011$ (10% off, from Gamma model of T_{avg}).

4 The Remaining 10%: Mature $T_{\text{avg}}(M)$

Open: Exact $T_{\text{avg}}(M)$ at Mature Nodes

The 10% gap between the self-consistency prediction and measurement traces directly to the Gamma distribution model for the KL field. The Gamma fit (shape $a = 9$, scale 0.015) was calibrated on early-run statistics; in the mature V20.3 regime, the KL distribution is more concentrated (crystallised nodes have lower mean KL), making $P(D_{\text{KL}} > K^*\sqrt{M})$ smaller and $T_{\text{avg}}(M)$ larger.

To close the proof:

1. Measure $T_{\text{avg}}(M)$ directly from the instrumented V20.3 by recording inter-event times per mass quartile (already done: $T_{\text{med}} \in \{20, 29, 45, 50\}$ for $M \in \{1.15, 1.38, 1.60, 1.81\}$).
2. Use measured $T_{\text{avg}}(M)$ to compute $p_{\text{ev}}(M) = \text{Poi}(\lambda)/T_{\text{meas}}(M)$ and the resulting $\langle 1/(1+M)\rangle_{\text{ev}}$ and $\langle\ln(1+M)\rangle_{\text{ev}}$.
3. Verify the self-consistency equation holds with measured T_{avg}.

This verification is one additional calculation. If it closes to $< 1\%$, the bracket identity is proved conditional on $T_{\text{avg}}(M)$ measurements.

5 Final Status Table

Result	Status	Note
bracket $= \kappa/\ln 2$ (numerical)	Confirmed	0.23%
$\lvert\beta\rvert = d_s/(2\ln 2)$ ($n = 8$)	Confirmed	3.4% finite-N
$f_{12} = d_s/2$ (accidental at $n = 8$)	Derived	Kuramoto + $d_s = 3$
$p_{\text{ev}}(M)$ form	Derived	Poisson/T_{avg}
$n = 2^{d_s}$ theorem	Refuted	$n = 4$ gives $d_s = 3.58 \neq 2$
$\lvert\beta\rvert = d_s/(2\ln 2)$ universal	Refuted	Fails at $n = 4$
Self-consistency equation	Confirmed	10% Gamma model error
Exact $T_{\text{avg}}(M)$ derivation	Open	Key remaining step
Full analytic proof of bracket	Open	Conditional on T_{avg}

Honest Maturity Assessment

Instability #21 for $n = 8$, $d_s = 3$:

- Numerically closed (0.23%). ✓
- Analytically 90% closed. All structure derived; one numerical input ($T_{\text{avg}}(M)$ in mature regime) needed for the final 10%.
- Not universal: formula is specific to $n = 8$ (where $f_{12} = d_s/2$ accidentally).

Maturity: **7.3/8**. The cascade is honest about what it knows and what it does not.

The formula held at $n = 8$ because, at three spatial dimensions, the half of the phase tension equals the half of the spectral count. At $n = 4$, the count changed but the tension did not. This is not a flaw in the formula. It is information about what the formula knows: it knows $d_s = 3$, and it knows Kuramoto, and the two happen to agree at $n = 8$. Whether they must agree is the question that separates a law of three dimensions from a law of the universe.